When imaging Gargantua in this chapter, I treat it as a fast-spinning black hole, as it must be to produce the extreme loss of time that the *Endurance*'s crew experience relative to Earth (Chapter 6). However, for fast spin, a mass audience could be confused by the flattening of the left edge of Gargantua's shadow (Figure 8.1) and by some peculiar features of the star streaming and the accretion disk, so Christopher Nolan and Paul Franklin chose a smaller spin, 60 percent of the maximum, for their Gargantua images in the movie. See the last section in Chapter 9.

Warning: The explanations in the following three sections may require a lot of thought; you can skip them without losing pace with the rest of the book. Not to worry!

The Shadow and Its Ring of Fire

The shell of fire (Chapter 6) plays a key role in producing Gargantua's shadow and the thin ring of fire alongside it. The shell of fire is the purple region surrounding Gargantua in Figure 8.2, and it contains nearly trapped photon orbits (light rays) such as the one in the upper right inset.[20]

Suppose you are at the location of the yellow dot. The white light rays $\mathcal{A}$ and $\mathcal{B}$ and others like them bring you the image of the ring of fire, and the black light rays $\mathcal{A}$ and $\mathcal{B}$ bring you the image of the shadow's edge. For example, the white ray $\mathcal{A}$ originates at some star far from Gargantua, it travels inward and gets trapped on the inner edge of the shell of fire in Gargantua's equatorial plane, where it flies round and round, driven by the whirl of space, and then escapes and comes to your eyes. The black ray also labeled $\mathcal{A}$ originates on Gargantua's event horizon, it travels outward and gets trapped on that same inner edge of the shell of fire, where it goes round and round, then escapes and